INTERNATIONAL UNIVERSITY – VIETNAM NATIONAL UNIVERSITY – HCMC

LINEAR ALGEBRA – MIDTERM

Semester 1, 2026–27 • Duration: 90 minutes

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Instructions:

- You can use two A4 sheets of notes. No electronic devices (including scientific calculator) are allowed.
- Each question carries 20 points.

1. Find 2×2 matrices A, B such that

$$A + B^T = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \qquad A^T - B = \begin{bmatrix} 1 & -3 \\ 1 & -1 \end{bmatrix}.$$

2. Consider the system

$$\begin{cases} x + 2y = 1, \\ y - 2z = -1, \\ x + y + (a^2 - a)z = a. \end{cases}$$

For which values of a does the system have infinitely many solutions? For which values of a does it have no solution?

3. Compute the determinant of

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 8 \\ -1 & 1 & -1 \end{bmatrix}.$$

4. Show that $\det(A + AB) = \det(BA + A)$ for any $n \times n$ matrices A, B.

5. Let A be the matrix in Problem 3 and $B = \det(A^{-1}) \cdot (A^{-1})^T$. Find B^{-1}.

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PROPOSED SOLUTIONS TO MIDTERM EXAMINATION

LINEAR ALGEBRA, October 30, 2026 • 2026 – 2027, Semester 1

Question 1.

Let $A = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$ and $B = \begin{bmatrix} y_1 & y_2 \\ y_3 & y_4 \end{bmatrix}$. Collectively, we derive the transpose of both matrices.

$$A^T = \begin{bmatrix} x_1 & x_3 \\ x_2 & x_4 \end{bmatrix}; \qquad B^T = \begin{bmatrix} y_1 & y_3 \\ y_2 & y_4 \end{bmatrix}$$

For each matrix equation, we essentially want to set up the corresponding system of linear equations via matrix operations, and then extract the sub-combinations (1) & (5), (2) & (7), (3) & (6), and (4) & (8), which can be solved for the variables x and y.

- $A + B^T = \begin{bmatrix} x_1 + y_1 & x_2 + y_3 \\ x_3 + y_2 & x_4 + y_4 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \Longrightarrow \begin{cases} x_1 + y_1 = 1 & (1) \\ x_2 + y_3 = -1 & (2) \\ x_3 + y_2 = 1 & (3) \\ x_4 + y_4 = 1 & (4) \end{cases}$

- $A^T - B = \begin{bmatrix} x_1 - y_1 & x_3 - y_2 \\ x_2 - y_3 & x_4 - y_4 \end{bmatrix} = \begin{bmatrix} 1 & -3 \\ 1 & -1 \end{bmatrix} \Longrightarrow \begin{cases} x_1 - y_1 = 1 & (5) \\ x_3 - y_2 = -3 & (6) \\ x_2 - y_3 = 1 & (7) \\ x_4 - y_4 = -1 & (8) \end{cases}$

- $\begin{cases} x_1 + y_1 = 1 \\ x_1 - y_1 = 1 \end{cases} \Longrightarrow \begin{cases} y_1 = 0 \\ x_1 = 1 \end{cases}$

- $\begin{cases} x_3 + y_2 = 1 \\ x_3 - y_2 = -3 \end{cases} \Longrightarrow \begin{cases} x_3 = -1 \\ y_2 = 2 \end{cases}$

- $\begin{cases} x_2 + y_3 = -1 \\ x_2 - y_3 = 1 \end{cases} \Longrightarrow \begin{cases} y_3 = -1 \\ x_2 = 0 \end{cases}$

- $\begin{cases} x_4 + y_4 = 1 \\ x_4 - y_4 = -1 \end{cases} \Longrightarrow \begin{cases} x_4 = 0 \\ y_4 = 1 \end{cases}$

$$\boxed{A = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}; \qquad B = \begin{bmatrix} 0 & 2 \\ -1 & 1 \end{bmatrix}}$$

Question 2.

Construct the augmented matrix $(A \mid b_n)$ and apply Gaussian elimination to uncover the pivots,

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 1 & -2 & -1 \\ 1 & 1 & a^2 - a & a \end{array}\right] \xrightarrow{R_3 = R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 1 & -2 & -1 \\ 0 & -1 & a^2 - a & a - 1 \end{array}\right] \xrightarrow{R_3 = R_3 + R_2} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & a^2 - a - 2 & a - 2 \end{array}\right]$$

Consequently, the linear system has infinitely many solutions if $a = 2$, making the third row becomes nill. Any value $a \neq 2$ will make the system unsolvable (no solutions).

In terms of $a = 2$, the solution set for this system is,

$$\boxed{\left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \begin{bmatrix} -4 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix} \right\}; \quad \forall t \in \mathbb{R}}$$

Question 3.

Perform elementary row operations on the determinant as following:

$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 4 & 8 \\ -1 & 1 & -1 \end{vmatrix} \xrightarrow[R_3 = R_3 + R_1 R_2]{R_2 = R_2 - 2R_1} \begin{vmatrix} 1 & 1 & 1 \\ 0 & 2 & 6 \\ 0 & 2 & 0 \end{vmatrix} = (-1)^2 \cdot \begin{vmatrix} 2 & 6 \\ 2 & 0 \end{vmatrix} = -12$$

Question 4.

Notice on the left-hand side, we can factor the matrix equation $(A + AB) = A(I_n + B)$, similar to the right hand side $(BA + A) = (B + I_n)A$.

Taking determinants and using the multiplicativity property,

$$\det(A(B + I_n)) = \det((B + I_n)A) \Longrightarrow \det(A) \cdot \det(B + I_n) = \det(B + I_n) \cdot \det(A)$$

Since the determinant is commutative $\det(AB) = \det(A) \cdot \det(B)$, as a result:

$$\det(B + I_n) = \det(I_n + B)$$

Therefore,

$$\det(AB + A) = \det(BA + A) \quad \forall A, B \in M_{n \times n} \qquad \square$$

Question 5.

Without computing the inverse of matrix A, we can still be able to find B^{-1} if know beforehand the property of inverse and transpose determinant

$$\det(A^{-1}) = \det(A)^{-1}; \quad (A^{-1})^T = (A^T)^{-1} \quad \Longrightarrow \quad B = \det(A)^{-1} \cdot (A^T)^{-1} = (\det(A) \cdot A^T)^{-1}$$

Taking the inverse on both sides, we obtain the B^{-1}

$$\boxed{B^{-1} = \det(A) \cdot A^T = -12 \cdot \begin{bmatrix} 1 & 2 & -1 \\ 1 & 4 & 1 \\ 1 & 8 & -1 \end{bmatrix} = \begin{bmatrix} -12 & -24 & 12 \\ -12 & -48 & -12 \\ -12 & -96 & 12 \end{bmatrix}}$$